



Cloud Security

(CST-109)

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Assignment

	Assignment 1: Load Balancer System	Assignment 2: Distributed File System
Goal	Efficiently distribute requests	Efficiently store & retrieve data/files
Focus	Request routing	Data storage & management
Main Problem	Which server should handle a request?	Where is the data stored and how to retrieve it?
Client Interaction	Sends many requests (up to 1M)	Uploads & downloads files
Servers Do	Process requests (stateless)	Store file chunks (stateful)
Key Component	Load Balancer	Metadata Server
Data Handling	No data storage	File chunking + storage
Replication	Not required (optional concept)	Mandatory (for fault tolerance)
Failure Handling	Redirect traffic to alive servers	Retrieve data from replicas
Evaluation Metric	Load distribution, response time	Data availability, reliability
Cloud Platforms	Private	Private

Assignment

	Assignment 3: Secure Cloud Application Architecture
Task	Objective
Task 1	Secure Architecture Design
Task 2	Authentication & Authorization
Task 3	Threat Modeling
Task 4	Risk Mitigation
Task 5	Resilience Evaluation
Cloud Platforms	Private

Cloud Computing

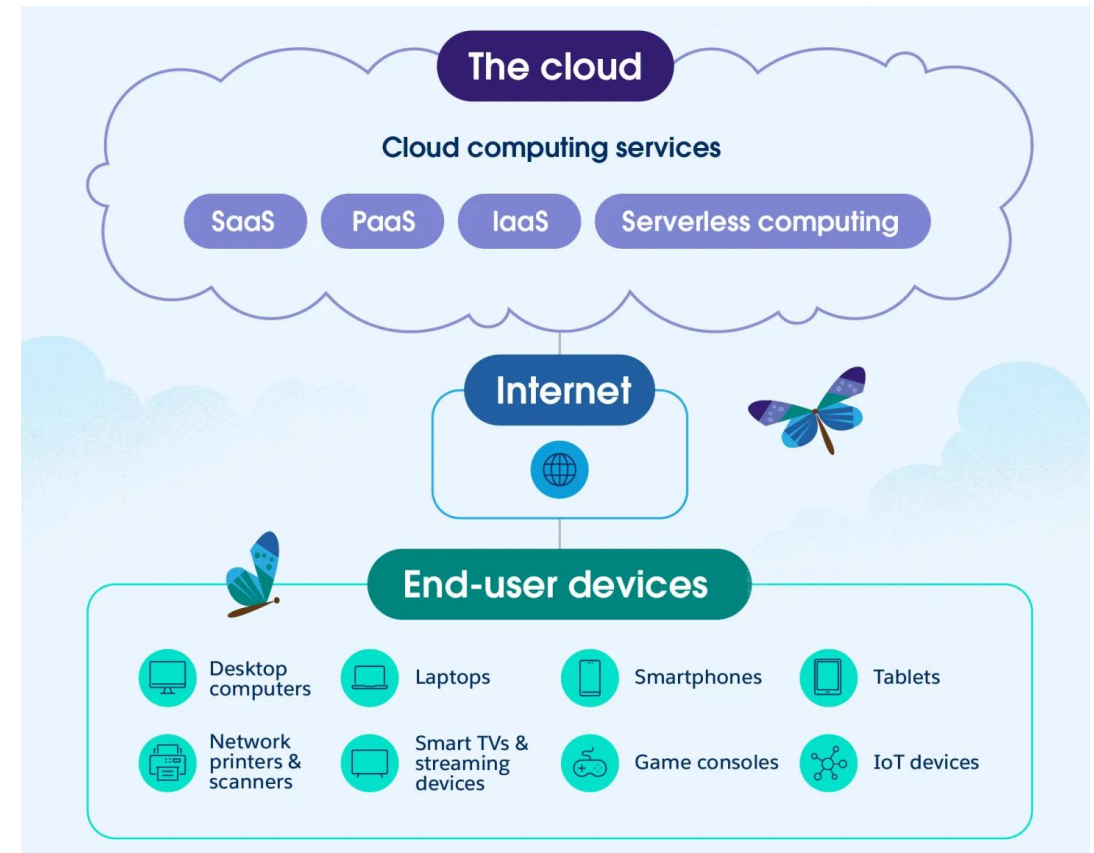
Cloud computing is the on-demand delivery of computing resources like servers, storage, databases, networking, and software over the internet

Private cloud

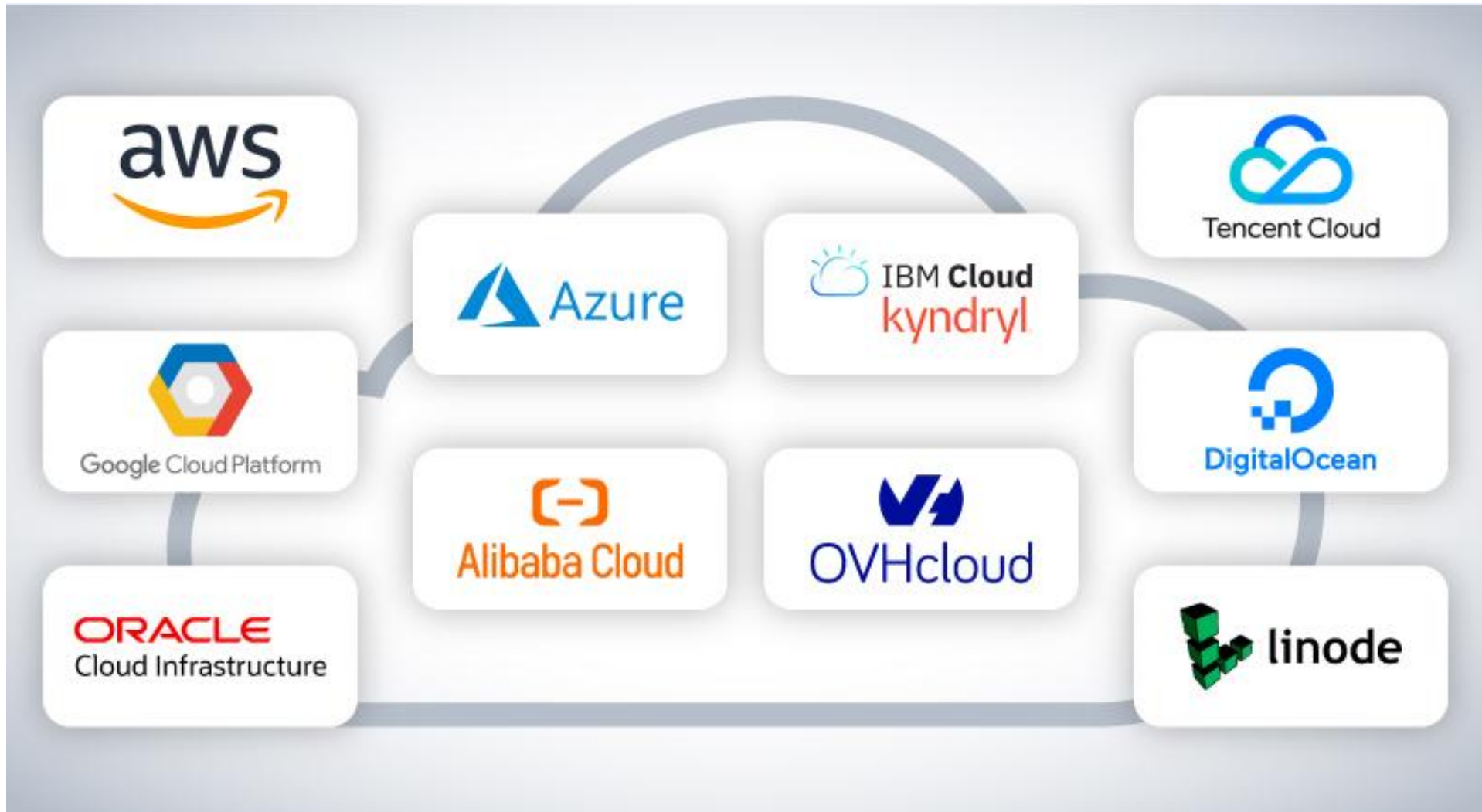
In this model, the service and infrastructure is dedicated solely to one organization

Public cloud

In a public cloud, services and infrastructure are hosted by third-party cloud providers and shared among multiple customers.



Cloud Platforms



Introduction to AWS Cloud Platform



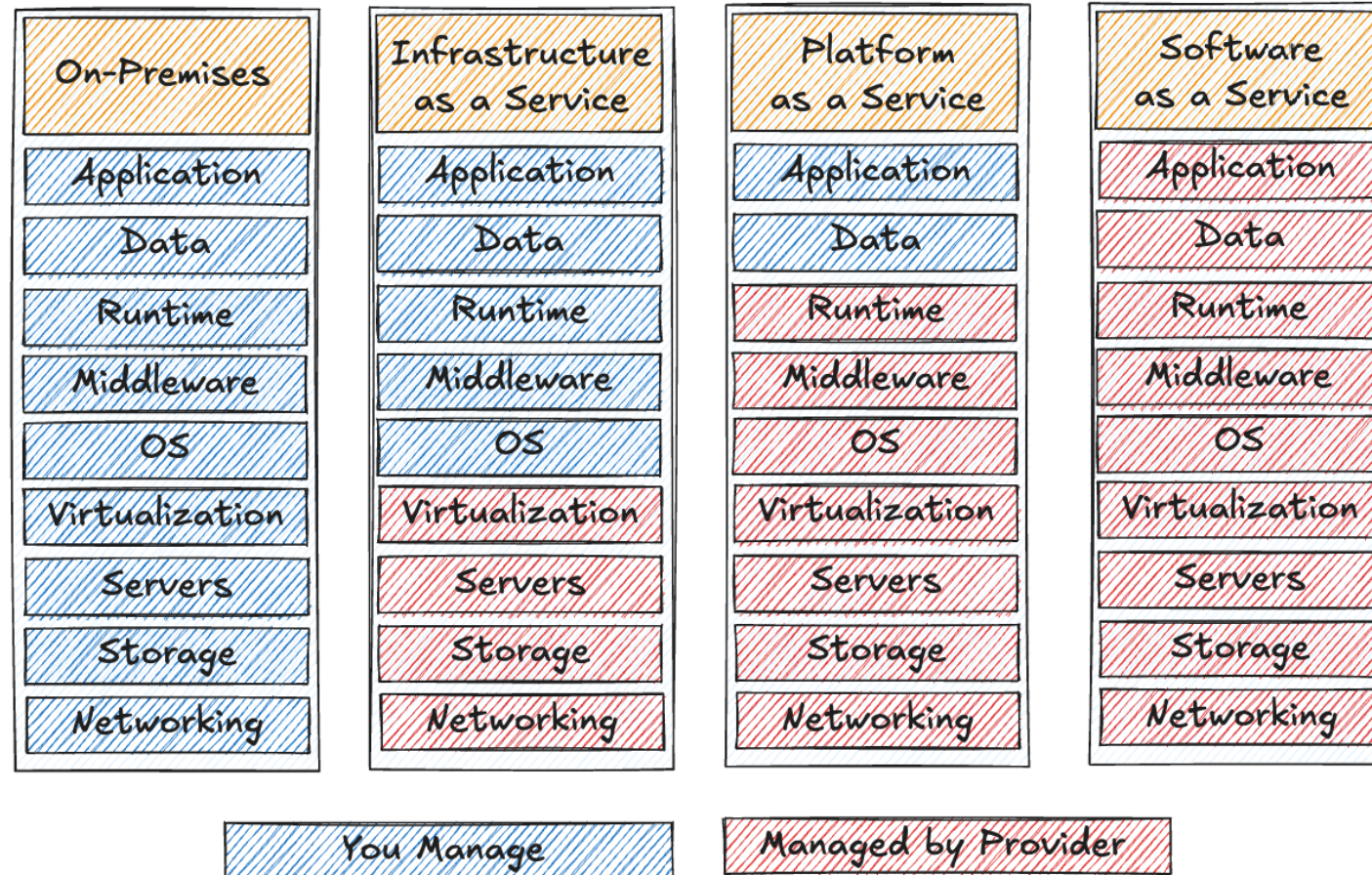
- ✓ **Create Free Login**
- ✓ **Try Creating Free Amazon EC2 virtual machine**
- ✓ **Access it from your Windows/ Linux**

[**Demo**](#)

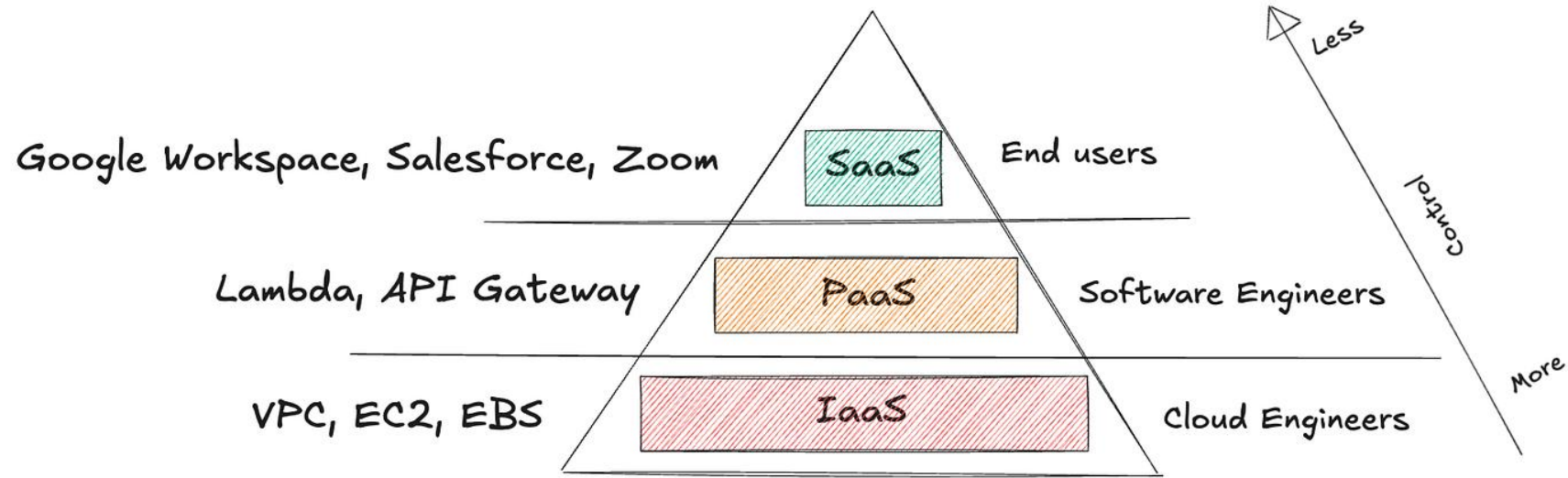
<https://www.youtube.com/watch?v=6bb-Oqbl8-E>

Introduction to AWS Cloud Services

AWS Cloud Service Models



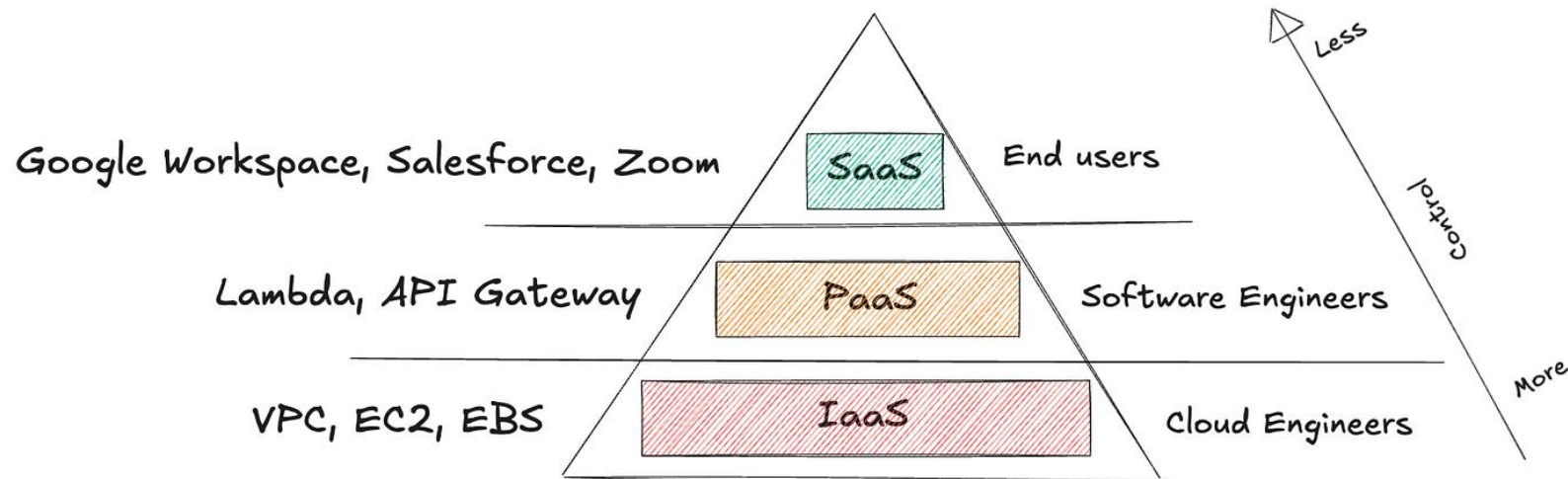
AWS- IaaS



- ✓ **Virtual Private Cloud (VPC):** Isolated network for managing IP ranges, subnets, and access control.
- ✓ **Elastic Compute Cloud (EC2):** Virtual servers to run applications.
- ✓ **Elastic Block Storage (EBS):** Persistent storage for EC2 instances.

Demo

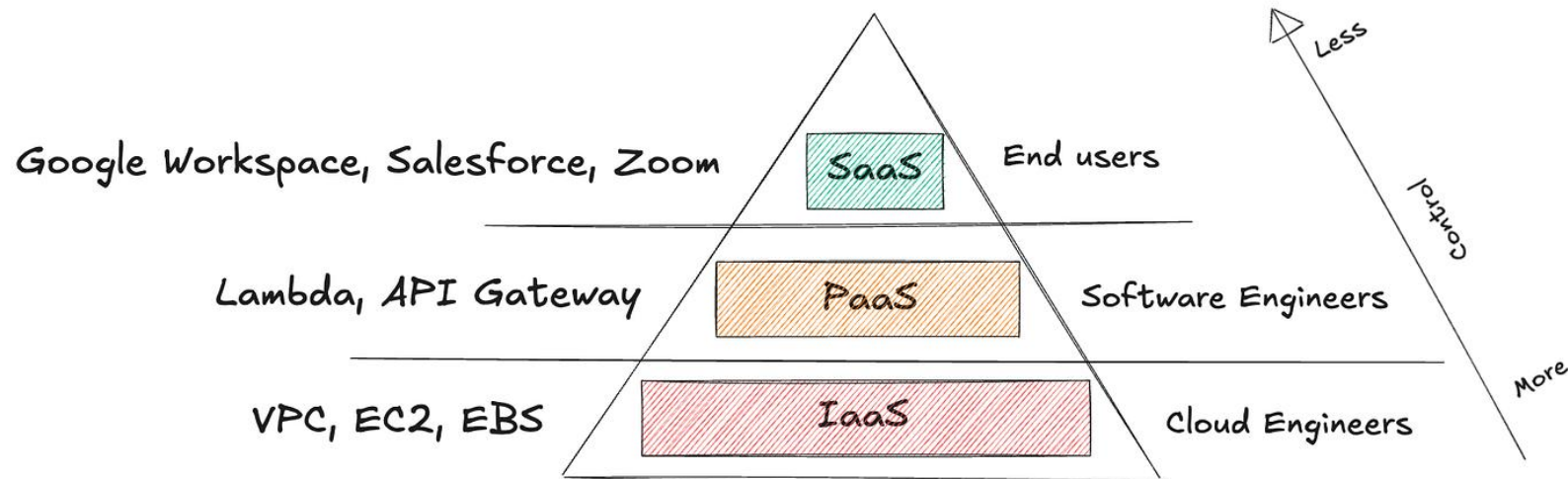
AWS- PaaS



- ✓ **Elastic Beanstalk:** A fully managed service that makes it easy to deploy and scale web applications and services.
- ✓ **RDS (Relational Database Service):** A managed service for setting up, operating, and scaling databases without having to worry about database maintenance, patching, or backups.
- ✓ **Lambda:** A serverless computing service that lets you run code in response to events without provisioning or managing servers.
- ✓ **API Gateway:** A fully managed service that makes it easy to create, publish, and manage APIs at scale.

Demo

IaaS



- Google Workspace:** A cloud-based suite of collaboration tools like Gmail, Calendar, Meet, Chat, Drive, Docs, Sheets, and Slides — a great choice for team collaboration.
- Zoom:** A communication platform that can be used for video conference meetings, audio conferences, webinars, and live chat.

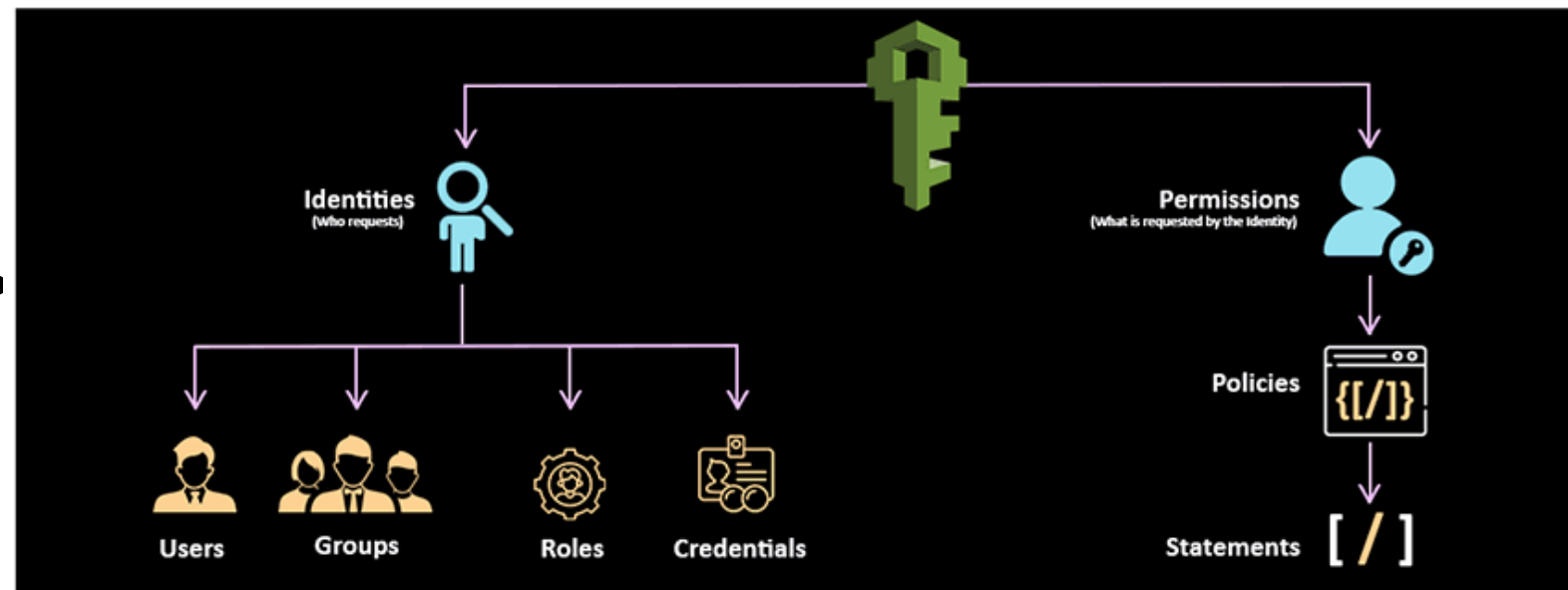
Introduction to AWS Security Tools



Introduction to IAM (Identity and Access Management)

- ✓ A core **secure access service** used in cloud platforms like AWS
- ✓ You should not use root user for using AWS account

ROOT USER →
**Controlling permission
of the AWS account**



Introduction to IAM (Identity and Access Management)

MFA (Multi-Factor Authentication)

Login requires **more than just a password**.

Even if someone steals your password, they **still cannot log in**.

The diagram illustrates the MFA process. On the left, the 'Sign in as IAM user' page shows fields for 'Account ID (12 digits) or account alias', 'IAM user name', and 'Password'. Below these is a 'Remember this account' checkbox and a 'Sign in' button. An orange arrow points to the right, where the 'Multi-factor Authentication' page is shown. This page prompts the user to 'Enter an MFA code to complete sign-in.' and features an 'MFA Code:' input field with the value '578126', a 'Submit' button, and a 'Cancel' link.

The screenshot shows the 'Multi-factor authentication (MFA) (0)' settings in the AWS IAM console. At the top right are buttons for 'Remove', 'Resync', and 'Assign MFA device'. The text below states: 'Use MFA to increase the security of your AWS environment. Signing in with MFA requires an authentication code from an MFA device. Each user can have a maximum of 8 MFA devices assigned. [Learn more](#)'. Below this is a table with columns: 'Type', 'Identifier', 'Certifications', and 'Created on'. The table is currently empty, with a message below it: 'No MFA devices. Assign an MFA device to improve the security of your AWS environment.' At the bottom right of the table area is an 'Assign MFA device' button.

JWT (JSON Web Token) is a **secure token** given to a user after login.

Introduction to IAM (Identity and Access Management)

Fine-Grained Authorization and Access Control

Give **very specific permissions** to users
- down to individual actions and resources.

Manages Users and Their Access

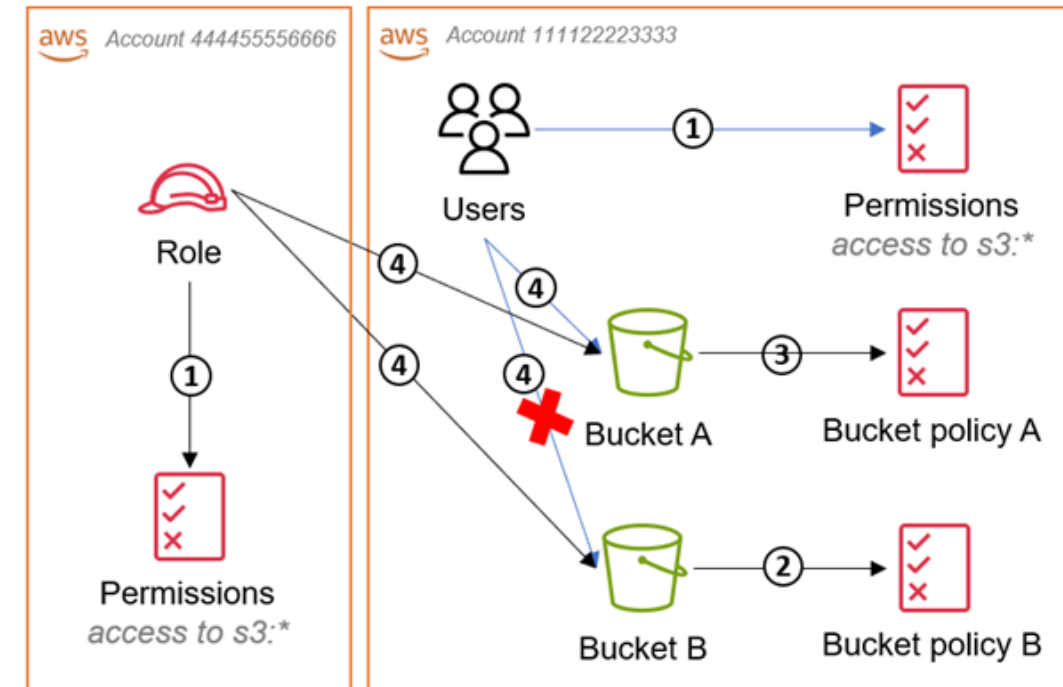
Create users: Alice, Bob, Admin

Each user gets **unique credentials and permissions**.

Manage Roles and Permissions

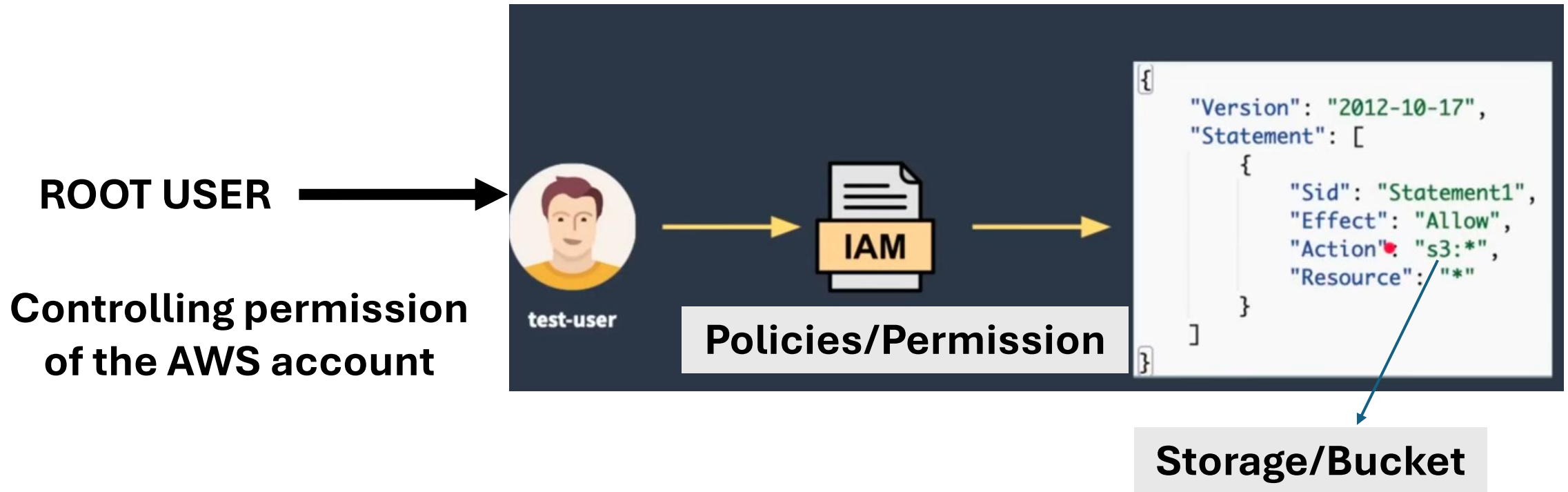
Role: EC2-Admin → can start/stop virtual machines

Role: S3-ReadOnly → can only view storage data



Role-Based Access Control (RBAC)

USE CASE



<https://www.youtube.com/watch?v=bO25vbkoJlA>

Demo



Thank You

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